

by induction on i , we prove the proposition, $([d_0, p_0], w, \gamma_1) \vdash_{P'}^i ([d, p], \epsilon, \gamma)$ implies $\delta_D(d_0, w) = d$ and $(p_0, w, \gamma_1) \vdash_P^i (p, \epsilon, \gamma) \text{ ——— (1)}$

(we intend to show whenever the LHS of (1) is true, the RHS of (1) holds)

basis: for $i = 0$ (no transitions applied), the above becomes, $([d_0, p_0], \epsilon, \gamma_1) \vdash_{P'}^{i=0} ([d_0, p_0], \epsilon, \gamma_1)$ implies $\delta_D(d_0, \epsilon) = d_0$ and $(p_0, \epsilon, \gamma_1) \vdash_P^{i=0} (p_0, \epsilon, \gamma_1)$

(since the LHS of (1) holds, the RHS of "implies" here proves the RHS of (1) holds when $i = 0$)

induction hypothesis (up to $i - 1$ steps, the following is assumed): $([d_0, p_0], x, \gamma_1) \vdash_{P'}^{i-1} ([d', p'], \epsilon, \gamma)$ implies $\delta_D(d_0, x) = d'$ and $(p_0, x, \gamma_1) \vdash_P^{i-1} (p', \epsilon, \gamma)$

induction step: $([d_0, p_0], w = xa, \gamma_1) \vdash_{P'}^{i-1} ([d', p'], a, \beta) \vdash_{P'} ([d, p], \epsilon, \gamma)$

from the induction hypothesis, after $i - 1$ steps, we know $\delta_D(d_0, x) = d'$ and $(p_0, x, \gamma_1) \vdash_P^{i-1} (p', \epsilon, \gamma)$ ——— (2)

since $\delta_{P'}$ definition yielded the last step, we know (due to iff in the definition of $\delta_{P'}$), $\delta_D(d', a) = d$ and $(p', a, \beta) \vdash_P (p, \epsilon, \gamma) \text{ ——— (3)}$

combining (2) and (3), lead to the RHS of implies of (1)